

260420 C++ 作業

Ch11 結構與其它資料型態

目錄

其他文章	1
Question 2	2
Question 7	4
Question 9	6

其他文章

在 union 裡面，所有成員都共享相同的內存，所以設置一個成員會影響所有成員。這可以用來節省內存，但這意味著你一次只能使用其中一個成員，並且使用一個會使其他成員失效。總的來說，這不太理想，並且可能導致各種各樣的錯誤。

```
union {  
    int a;  
    int b;  
}
```

如果我更改 a，b 也會更改。

我們早就過了需要數內存的年代，所以現在的 unions 通常保留用於處理一些非常具體的低級數據打包問題。總的來說，除非你真的知道你需要使用它們，否則不要使用它們。

[Reddit: C - Is it better to use struct or union? When should I use which?](#)

Question 2

```
1 #include <iostream>
2 #include <iomanip>
3 using namespace std;
4
5 struct Date
6 {
7     int year;
8     int month;
9     int day;
10 };
11
12 int main()
13 {
14     Date input;
15     cout << "輸入年份 yyyy: "; cin >> input.year;
16     cout << "輸入月份 mm: "; cin >> input.month;
17     cout << "輸入日期 dd: "; cin >> input.day;
18
19     cout << setfill('0') << setw(2) << input.month << "/"
20          << setfill('0') << setw(2) << input.day << "/"
21          << input.year << "\n";
22
23     return 0;
24 }
25
```

CP Editor-gcFg

New Tab Split View

輸入年份 yyyy: 2026
輸入月份 mm: 04
輸入日期 dd: 20
04/20/2026

Program finished with
Press any key to exit

```
1 #include <iostream>
2 #include <iomanip>
3 using namespace std;
4
5 struct Date
6 {
7     int year;
8     int month;
9     int day;
```

```
10 };
11
12 int main()
13 {
14     Date input;
15     cout << "輸入年份 yyyy: "; cin >> input.year;
16     cout << "輸入月份 mm: "; cin >> input.month;
17     cout << "輸入日期 dd: "; cin >> input.day;
18
19     cout << setfill('0') << setw(2) << input.month << "/"
20         << setfill('0') << setw(2) << input.day << "/"
21         << setfill('0') << setw(4) << input.year << "\n";
22
23     return 0;
24 }
25
```

Question 7

```
1 #include <iostream>
2 using namespace std;
3
4 struct Info
5 {
6     string name;
7     int ID;
8     int gender;
9     union
10    {
11        bool has_served;
12        int typing_speed;
13    };
14 };
15
16 int main()
17 {
18     Info employee;
19
20     cout << "姓名: ";      getline(cin, employee.name);
21     cout << "人事代號: ";  cin >> employee.ID;
22     cout << "性別 (1)男 (2)女: "; cin >> employee.gender;
23     if (employee.gender == 1) { cout << "是否役畢 (1)是 (0)否: "; cin >> employee.has_served; }
24     else if (employee.gender == 2) { cout << "每分鐘中打速度: "; cin >> employee.typing_speed; }
25
26     cout << "姓名: " << employee.name << "\n";
27     cout << "人事代號: " << employee.ID << "\n";
28     if (employee.gender == 1) { cout << "是否服役: " << employee.has_served << "\n"; }
29     else if (employee.gender == 2) { cout << "每分鐘中打速度: " << employee.typing_speed << "\n"; }
30
31     return 0;
32 }
33
```

姓名: Tippi Hong
人事代號: 113005
性別 (1)男 (2)女: 1
是否役畢 (1)是 (0)否: 1
姓名: Tippi Hong
人事代號: 113005
是否服役: 1

Program finished with exit code
Press any key to exit

```
1 #include <iostream>
2 using namespace std;
3
4 struct Info
5 {
6     string name;
7     int ID;
8     int gender;
```

```
9     union
10     {
11         bool has_served;
12         int typing_speed;
13     };
14 };
15
16 int main()
17 {
18     Info employee;
19
20     cout << "姓名: ";           getline(cin, employee.name);
21     cout << "人事代號: ";       cin >> employee.ID;
22     cout << "性別 (1)男 (2)女: "; cin >> employee.gender;
23     if      (employee.gender == 1) { cout << "是否役畢 (1)是 (0)否: ";   cin >> employee.has_served;   }
24     else if (employee.gender == 2) { cout << "每分鐘中打速度: ";         cin >> employee.typing_speed; }
25
26     cout << "姓名: " << employee.name << "\n";
27     cout << "人事代號: " << employee.ID << "\n";
28     if      (employee.gender == 1) { cout << "是否服役: " << employee.has_served << "\n";   }
29     else if (employee.gender == 2) { cout << "每分鐘中打速度: " << employee.typing_speed << "\n"; }
30
31     return 0;
32 }
33
```

Question 9

```
4 enum Month
5 {
6     Jan = 1,
7     Feb,
8     Mar,
9     Apr,
10    May,
11    Jun,
12    Jul,
13    Aug,
14    Sep,
15    Oct,
16    Nov,
17    Dec
18 };
19
20 int main()
21 {
22     cout << "請輸入欲查詢的月份(1~12): ";
23     int n; cin >> n;
24
25     switch (n)
26     {
27         case Jan: cout << n << "月 = January"; break;
28         case Feb: cout << n << "月 = February"; break;
29         case Mar: cout << n << "月 = March"; break;
30         case Apr: cout << n << "月 = April"; break;
31         case May: cout << n << "月 = May"; break;
32         case Jun: cout << n << "月 = June"; break;
33         case Jul: cout << n << "月 = July"; break;
34         case Aug: cout << n << "月 = August"; break;
35         case Sep: cout << n << "月 = September"; break;
36         case Oct: cout << n << "月 = October"; break;
37         case Nov: cout << n << "月 = November"; break;
38         case Dec: cout << n << "月 = December"; break;
39         default: cout << "Error"; break;
40     }
41     cout << endl;
```

請輸入欲查詢的月份(1~12): 5
5月 = May

Program finished with exit c
Press any key to exit

```
1 #include <iostream>
2 using namespace std;
3
4 enum Month
```

```
5 {
6     Jan = 1,
7     Feb,
8     Mar,
9     Apr,
10    May,
11    Jun,
12    Jul,
13    Aug,
14    Sep,
15    Oct,
16    Nov,
17    Dec
18 };
19
20 int main()
21 {
22     cout << "請輸入欲查詢的月份(1~12): ";
23     int n;  cin >> n;
24
25     switch (n)
26     {
27         case Jan:  cout << Jan << "月 = January";    break;
28         case Feb:  cout << n << "月 = February";      break;
29         case Mar:  cout << n << "月 = March";          break;
30         case Apr:  cout << n << "月 = April";          break;
31         case May:  cout << n << "月 = May";            break;
32         case Jun:  cout << n << "月 = June";           break;
33         case Jul:  cout << n << "月 = July";           break;
34         case Aug:  cout << n << "月 = August";         break;
35         case Sep:  cout << n << "月 = September";      break;
36         case Oct:  cout << n << "月 = October";        break;
37         case Nov:  cout << n << "月 = November";       break;
38         case Dec:  cout << n << "月 = December";      break;
39         default:   cout << "Error";                   break;
40     }
```

```
41     cout << endl;  
42  
43     return 0;  
44 }  
45
```